ORAL HISTOLOGY AND EMBRYOLOGY

1. Which period of pre-natal development does fertilization and implantation happen?
A. Embryonic Period
B. Proliferative Period
C. Fetal Period
D. A and B
E. None of the above

2. Which of the following is true regarding implantation?
A. Occurs by the end of the first week
B. Trophoblast cells invade endometrial epithelium
C. Trophoblast produces hCG
D. A and B
E. All of the above

3. The following are ectopic sites of implantation except:
A. Uterine tube
B. Cervical opening of the uterus
C. Uterus
D. Ovary
E. Peritoneal cavity lining

4. After what stage of zygote development does the zona pellucida disintegrate?
A. Zygote
B. Blastomere
C. Blastocyst
D. Late morula
E. Early morula

Which of the following best describes the decidual reaction of embryogenesis?
A. Breakdown of the zona pellucida
B. The first zygotic cleavage
C. Invasion of the trophoblast in the endometrium
D. Release of an ovum from the ovary
E. Enlargement of the endometrium with lipid and glycogen accumulation

6. Which of the following is NOT an endodermal derivative to GI tract?
A. Liver
B. Esophagus
C. Oropharynx
D. Pancreas

7. Which of the following germ layer derivatives comprise the brain and spinal cord?
A. Mesoderm
B. Endoderm
C. Ectoderm
D. All of the above
E. None of the above

8. The neural plate invaginates to form neural groove with neural folds on each side on day 18 of embryonic life. Neuroectoderm gives rise to CNS.
A. First statement is true. Second statement is false,
B. First statement is false. Second statement is true.
C. Both statements are true.
D. Both statements are false.

9. A cranial structure is close to which part of the embryo?
A. posterior wall
B. Inferior end
C. superior end
D. anterior wall

10. Primitive oral cavity is ___
A. Dental Papilla
B. Tooth bud
C. Dental Lamina
D. Oral stomodeum

11. The embryological stomodeum is located at the:
A. Ventral side of the cephalic end of embryo.
B. Dorsal side of the cephalic end of embryo.
C. Ventral side of the caudal end of embryo.
D. Dorsal side of the caudal end of embryo.

12. The buccopharyngeal membrane is lined by:
A. Ectoderm
B. Mesoderm
C. Endoderm
D. A and C
E. B and C

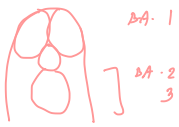
13. Which branchial arch has its derivatives innervated by CN V?
A. I
B. II
C. III
D. IV

14. The second branchial arch gives rise to muscles of
A. mastication
B. the pharynx
C. the oropharynx
D. facial expression

15. Which of the following is the nerve of the fourth branchial arch?
A. Facial Nerve
B. Trigeminal Nerve
C. Vagus
D. Glossopharyngeal

16. The external auditory meatus is derived from:
A. First pharyngeal pouch
B. Second pharyngeal pouch termination
C. First pharyngeal cleft
D. First pharyngeal arch

17. All of the following structures are concerned with the development of the tongue except
A. copula
B. placodes
C. tubercular impar
D. second branchial arch
E. third pharyngeal arch



18. Vagus nerve supplies the pharyngeal and laryngeal muscles, because it develops from ___
A. 2nd branchial arch
B. 6th branchial arch
C. 4th Branchial arch
D. Two of the above

19. In the development of the face, what brings about a separation between lips and cheeks externally and the jaw structures internally?
A. Tuberculum impar
B. Buccopharyngeal membrane
C. Vestibular lamina
D. Dental lamina
E. Lateral lingual swellings

20. The bridge and tip of the nose are derived from the
A. frontonasal elevation
B. mandibular process
C. medial nasal elevation
D. lateral nasal elevation

21. Bilateral cleft lip results from failure to merge of the
A. maxillary process and the medial nasal elevation
B. two medial nasal elevation
C. medial and lateral nasal elevation
D. intermaxillary segment and the lateral nasal elevation

22. All of the following are affected in complete unilateral palatal clefting except:
A. Lip
B. Primary palate
C. Alveolar process
D. Inferior nasal turbinate
E. Secondary palate

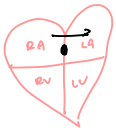
23. In human embryo, during the 3th week of development, the mandibular arch divides to form the:
A. Mandibular process
B. Maxillary process
C. Frontonasal process
D. A and B
E. All of the above

24. Which of the following is FALSE regarding the primary palate?
A. It cannot be involved in facial clefting.
B. It contains the incisive foramen.
C. It usually contains teeth.
D. It is mesenchymal in origin.
E. It is formed by fusion of two median nasal processes.



25. A woman with cleft lip is concerned about the risk of occurrence of that defect in her future child. She can be counseled that ___
A. Cleft lip would only occur if both parents were affected
B. There are empiric data on the recurrence risk of cleft lip
C. Cleft lip is not thought to have a genetic basis
D. Only a male would be a risk of inheriting cleft lip.

26. Regarding the fetal circulation, oxygenated blood bypasses the pulmonary circulation by way of which communication?
A. Left atrium to left ventricle
B. Left atrium to right atrium
C. Right atrium to right ventricle
D. Right atrium to left atrium
E. Left atrium to left ventricle



27. Which of the following structures of the adult heart is a remnant of the ductus venosus in the fetal heart?
A. Crista terminalis
B. Fossa ovalis
C. Sulcus terminalis
D. Ligamentum venosum
E. Ligamentum arteriosum

28. Regarding the fetal circulation, which of the following vasculature serves as a conduit between the right atrium and left atrium?
A. Ductus venosus
B. Umbilical arteries
C. Ductus arteriosus
D. Foramen ovale
E. Umbilical vein



29. Ligamentous remnants of the fetal circulatory system persisting in the adult include the ligament ___
I. Nuchae
II. venosum
III. arteriosum
IV. teres of the liver
V. teres of the uterus
A. II,III,IV
B, I,II,V
C. I,II,III
D. I,IV,V

30. During odontogenesis, which of the following cells determine the type of tooth to be formed?
A. Mesoderm
B. Mesenchyme
C. Ectomesenchyme
D. Endoderm
E. Ectoderm

31. The oral epithelium, arises from _____.
A. Endoderm
B. Neural crest cells
C. Ectoderm
D. Mesoderm

32. Tooth enamel is derived from ectoderm.
A. True
B. False, it is from mesoderm
C. False, it is from endoderm
D. B and C

33. What induces that oral epithelium to proliferate, eventually forming dental lamina?
A. Neuroepithelial cells
B. Neural Crest cells
C. A and B
D. None of the above

34. Promordia of dental pulp is ___
A. Dental lamina
B. Tooth Buds
C. Dental Papilla
D. Dental Sac

35. The cementum is derived from the ___
A. dental follicle
B. outer dental epithelium
C. epithelial root sheath
D. dental papilla

36. Formation of Tooth Germ happens in the Cap Stage, it consists of:
A. Enamel organ only
B. Dental papilla only
C. Dental sac only
D. All of the above

37. The correct order of tooth formation is _____.
A. Ameloblasts form, odontoblasts form, ameloblasts start to form enamel, odontoblasts start to form dentin
B. Ameloblasts, odontoblasts, odontoblasts start to form dentin, ameloblasts start to form enamel
C. Odontoblasts form, odontoblasts start to form dentin, ameloblasts form, ameloblasts start to form enamel
D. Ameloblasts form, ameloblasts start to form enamel, odontoblasts form, odontoblasts start to form dentin
E. Odontoblasts form, ameloblasts form, odontoblasts start to form dentin, ameloblasts start to form enamel

38. The _____ differentiates into odontoblasts.
A. Periphery of dental papilla in the bell stage
B. Inner enamel epithelium in the cap stage
C. Inner enamel epithelium in the bell stage
D. Core of the dental papilla in the cap stage
E. Periphery of dental papilla in the cap stage

39. The formative cells of the alveolar bone arise from the ____
A. inner enamel epithelium.
B. dental follicle.
C. dental papilla.
D. dental follicle and papilla.
E. inner enamel epithelium and dental papilla.

40. The dental follicle plays an important role in the formation of
1. the bone of the alveolar socket
2. enamel over the crown of the tooth
3. root dentin in the developing tooth
4. the fibrous periodontal ligament
5. cementum over the root of the tooth

A. 1
B. 1,4,5
C. 1,3,5
D. 2,4
E. 3,4

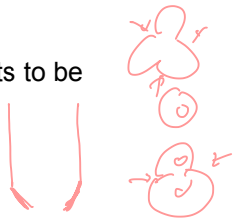
41. At what stage of development does the formation of tooth germ happen?
A. Initiation
B. Bud stage
C. Cap stage
D. Bell stage

42. When do odontoblasts begin to secrete dentin matrix?
A. During the cap stage
B. After the ameloblasts form enamel
C. Before the ameloblasts form enamel
D. After root formation begins
E. Before cap stage

43. Which of the following is FALSE regarding gubernacular cord?
A. It is found in the gubernacular canals
B. it is fibrous tissue with the remnants of the dental lamina
C. it is found in the labial of deciduous teeth
D. it is the connection between the succedaneous teeth and the lamina propria of the oral mucosa



44. What will determine the number of roots to be formed
A. Epithelial diaphragm
B. Hertwig's epithelial root sheath
C. Periodontal ligament
D. Cushion hammock ligament



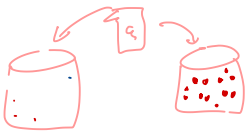
45. This is formed from the reduced enamel epithelium and the reduced oral epithelium forming a collar around the tooth
A. Junctional epithelium
B. Marginal gingiva
C. Free gingiva
D. Sulcular epithelium

46. Union of Outer and Inner enamel epithelium
A. Epithelial rest of Malassez
B. Junctional epithelium
C. Cervial loop
D. Reduced enamel epithelium
E. Nasmyth's membrane

47. Last product produced by ameloblasts
A. Primary enamel cuticle
B. Secondary enamel cuticle
C. Reduced enamel epithelium
D. Junctional epithelium

48. Which of the following has the HIGHEST inorganic composition?
A. Cementum
B. Dentin
C. Pulp
D. Enamel
E. Alveolar bone

49. Which of the following is usually absent in a slide prepared by decalcification?
A. Cellular cementum
B. Odontoblasts
C. PDL Fibers
D. Mature enamel



50. Appositional growth of enamel
A. 0.04 µm.
B.1mm.
C. 0.1 mm.
D. 4 µm.
E. 4mm.

51. Structural characteristics of enamel except:
A. Lines of retzius
B. Transverse striations
C. Granular layer of tomes
D. Hunter-Schreger bands

52. Hypocalcified enamel defects that can extend all the way to enamel surface
A. Transverse striations
B. Enamel lamellae
C. Bands of retzius
D. Dentinal tubules

53. The structural characteristic of enamel that promotes lateral spread of caries
A. Enamel lamellae
B. Enamel spindle
C. Enamel tufts
D. Enamel pearl



54. Enamel is harder than bone because it is tightly packed with smaller enamel hydroxyapatite crystals, four times smaller than those in bone. The enamel prisms form the structural foundation of enamel. The first statement is ___
a. incorrect and the second statement is correct
b. correct and the second statement is correct
c. correct and the second statement is incorrect
d. incorrect and the second statement is incorrect

55. Enamel is a highly mineralized structure containing approximately 95% of organic matter. The largest mineral constituents are hydroxyapatite crystals. The first statement is: ___
a. correct and the second statement is correct
b. incorrect and the second statement is incorrect
c. incorrect and the second statement is correct
d. correct and the second statement is incorrect

56. Enamel is semi permeable and allows fluoride ions to be absorbed in the hydroxyapatite crystals forming fluorapatite. The tooth becomes more resistant to bacteria producing acids because fluorapatite has a higher solubility product content than hydroxyapatite. the first statement is ___
A. Incorrect and the second statement is incorrect
B. Incorrect and the second statement is correct
C. Correct and the second statement is incorrect
D. correct and the second statement is correct

57. Which of the following statements about enamel is correct?
I. About 95% of enamel is inorganic
II. The major constituents as for dentin are calcium, phosphorus, carbonate and magnesium
III. The water content of enamel is estimated between 2-3%
IV. Amino acid make up of enamel protein does not identify it as keratin, the protein of other ectodermal structures.
V. Enamel is ectodermal in origin

A. I, II, III, IV, V
B. I, II, IV, V
C. I, II, III, V
D. II, IV, V

58. Which is a type of enamel defect that is strongly associated with amelogenesis imperfecta?
A. Enamel hypercalcification
B. Enamel hyperplasia
C. Enamel hypoplasia
D. Enamel hypocalcification

59. An elastic, avascular, mineralized tissue that is harder than bone but softer than enamel
A. Dentin
B. Cementum
C. Pulp
D. Periodontal Ligament
E. None of the above

60. How many percent of water does dentin have?
A. 1-5
B. less than 1
C. 5 - 10
D. 15 - 25

61. If several adjacent dentinal tubules become occluded with peritubular dentin, it takes on a glassy appearance and is called ____
A. Globular dentin
B. Sclerotic dentin
C. Dead Tracts
D. Interglobular dentin
E. None of the above



62. What line indicates the early variation in the secretory activity of dentin?
A. Salter's
B. Cement
C. Von' ebner's
D. Line of Retzius

63. The initial 150um of dentin is called _____
A. Mantle dentin
B. Circumpulpal dentin
C. Secondary dentin
D. Pre-dentin

64. Groups of necrotic odontoblastic process within dentinal tubules often caused by trauma
A. Globular dentin
B. Sclerotic dentin
C. Dead Tracts
D. Interglobular dentin
E. None of the above



65. Coronal dentinal tubules follow an S shape. Radicular tubules are generally straight.
A. First statement is true. Second statement is false,
B. First statement is false. Second statement is true.
C. Both statements are true.
D. Both statements are false.

66. The type of dentin form after root formation as a response to an outside or noxious stimulus or even therapeutic intervention is ___
A. Tertiary
B. Secondary
C. Intermediary
D. Primary

67. The type of dentin is formed as a response to a functional stimuli
A. Reparative
B. Tertiary
C. Secondary
D. Primary

68. What is the most likely cause if a patient complains of tooth sensitivity and clinical examination reveals areas of gingival recession exposing the roots of each teeth?
A. Fluids entering the lacunae of the cementum
B. Fluids entering the enamel rods
C. Movement of fluid in the sulcular epithelium
D. Movement of fluid in the dentinal tubules

69. As the dentin ages, which of the following is typically seen?
A. Increased dentinal tubule diameter
B. Decreased reparative dentin
C. Decreased dead tracts
D. Decreased sclerotic dentin
E. Increased deposition of peritubular dentin

70. The number/amount of the following structures can give an indication on the age of the tooth, except:
A. Enamel lamellae
B. Enamel tufts
C. Sclerotic dentin
D. Cellular cementum

71. Single layer of odontoblasts lining in pulp chamber
A. Pulp core
B. Cell rich zone
C. Cell free zone of Weil
D. Odontoblastic zone
E. None of the above

72. Functions of pulp except:
A. Formative
B. Nutritive
C. Sensory
D. Protective
E. All of the above

73. Which of the following portions of the tooth or its associated structures allows for orthodontic movement?
A. Enamel
B. Cementum
C. Dentin
D. Epithelial attachment
E. Alveolar bone

74. Cementum is a vascular tissue that covers radicular dentin. Composition most closely resembles bone.
A. First statement is true. Second statement is false,
B. First statement is false. Second statement is true.
C. Both statements are true.
D. Both statements are false.

75. Secretory activity of cementum
A. Salters line
B. Retzius line
C. Von ebners line
D. Cementing line

76. The type of cementum that is first laid down and directly covers the dentinal tubules is ___
A. Cellular intrinsic fiber cementum
B. Intermediate cementum
C. Acellular intrinsic fiber cementum
D. Cellular mixed fiber cementum

77. Which of the following statements is NOT true regarding cementum?
A. It has more mineral content than bone
B. It is continuously deposited at the apical portion
C. It has a component of the tooth and periodontium
D. It seals the ends of the dentinal tubules

78. Which of the following sequence is the correct order from the most to the least radiopaque?
A. Enamel, Bone, Dentin
B. Enamel, Dentin, Bone, Pulp
C. Dentin, Enamel, Bone
D. Enamel, Dentin, Bone

79. Which of the following statements is correct?
A. Bone and cementum do not resorb
B. Cementum is more resistant to resorption than bone
C. Bone and cementum resorb at the same rate
D. Bone is more resistant to resorption than cementum

80. Bundle bone is the type of bone that
1. lines the socket in teeth that have been subjected to stress
2. has been formed by endochondral ossification
3. anchors the Sharpey's fiblers of the periodontal ligament
4. is the first formed around the developing tooth
5. contains fewer matrix collagen fibrils than typical bone

A. 1
B. 1,2,5
C. 1,3,5
D. 2,4
E. 3,4

81. Growth line of alveolar bone
A. Salters line
B. Retzius line
C. Von ebners line
D. Cementing line

82. The periodontium is the attachment apparatus of the tooth. It consists of Cementum, Alveolar bone, Periodontal ligament and Dentin.
A. First statement is true. Second statement is false.
B. First statement is false. Second statement is true.
C. Both statements are true.
D. Both statements are false.

83. Principal collagen fibers that extends inter proximally over the alveolar crest from the cementum of one tooth to that of adjacent tooth.
A. Transseptal
B. Interradicular
C. Horizontal
D. Apical
E. Oblique



84. Principal collagen fibers of the PDL that are the most numerous
A. Transseptal
B. Interradicular
C. Horizontal
D. Apical
E. Oblique

85. Attached to the cementum just below the CEJ and running downward and outward to insert into the rims of alveolus
A. Transseptal
B. Interradicular
C. Horizontal
D. Apical
E. None of the above



86. The periodontal ligament is vital to the functional life of the tooth because it
I. furnishes an elastic sling for holding the tooth
II. provides a cellular source for new cementum and bone
III. allows for physiologic movement of the tooth
IV. contains nervous and vascular elements

A. I, II, III
B. I,III
C. I,II,IV
D. II, III,IV

87. The periodontal ligament contains collagen fibers that are inserted on one side in the cementum and on the other side in alveolar bone. the ends of these collagen fibers are sharpey's fibers
A. First statement is True, Second statement is False
B. Both statements are True
C. Both statements are False
D. Second statement is True, First Statement is False

88. Formative Cell of periodontal ligament
A. Odontoblast
B. Cementoblast
C. Fibroblast
D. Ameloblast

89. What happens when the teeth fails to grow?
A. The alveolar process fails to form
B. The alveolar process continues to grow
C. The alveolar process partially grow
D. The alveolar process excessively grow

90. The free gingival groove separates the gingiva from alveolar mucosa. The mucogingival junction separates the attached and unattached gingiva.
A. First statement is true. Second statement is false.
B. First statement is false. Second statement is true.
C. Both statements are true.
D. Both statements are false.

91. Projections that interdigitate with the epithelial ridges, creating a variably tortuous epithelial-connective tissue interface.
A. Stratum granulosum
B. Papillary layer of lamina propria
C. Stratum corneum
D. Reticular layer of lamina propria
E. Stratum spinosum

92. Gingival fiber group that extend from cementum over the alveolar crest, and turn apically to insert to periosteum of the buccal side of the alveolar bone.
A. Dentogingival fibers
B. Alveologingival fibers
C. Dentoperiosteal fibers
D. Circumferential fibers



93. Depression in interdental gingival connecting the buccal and lingual papilla
A. Gingival col.
B. Alveolar mucosa.
C. Attached gingiva.
D. Mucogingival junction.
E. Interdental papilla.

94. Derived from reduced enamel epithelium and oral epithelium
A. Oral sulcular epithelium
B. Junctional epithelium
C. Oral gingival epithelium
D. All of the above

95. Coronally bounded by free gingival groove and apically by mucogingival junction
A. Marginal gingiva
B. Interdental papilla
C. Alveolar mucosa
D. Attached gingiva

96. Change in position of tooth germ during bell stage of tooth development before they begin to erupt
A. Pre-eruptive phase
B. Pre-fuctional phase
C. Eruptive tooth movement
D. Post-eruptive tooth movement

97. Teeth that failed to erupt completely simply because of lack of eruptive force are termed as
A. Impacted teeth
B. Embedded teeth
C. Submerged teeth
D. Infra-verted teeth

98. Physiologic mesial drift is a tooth movement that
A. is required for pre-eruptive tooth movement
B. compensates for inter proximal tooth wear
C. compensates for occlusal tooth wear
D. Accommodates growing jaws

99. What is passive eruption?
A. Movement of tooth to compensate for proximal wear
B. Movement of tooth to compensate for occlusal wear
C. Gradual exposure of the crown and later of the root by gingival recession
D. Movement of the tooth towards the occlusal plane

100. Lack of eruption may be related to fusion of roots to bony sockets or to crown of permanent tooth
A. Impacted tooth
B. Ankylosis
C. Submerged tooth
D. Embedded tooth